

Almost a Policy: Transmission through Administrative Handoffs

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A policy reaches a recipient through administrative handoffs that transmit and measure an intervention. Each handoff is represented as a Markov transformation of an upstream experiment followed by information produced at the receiving stage. For a vector counterfactual, loss and acquisition ranks form a stage ledger. The loss rank equals the minimum score rank of an upstream channel that restores local identification. Restoring full-information precision requires rank $\text{rank}\{L^{1/2}(I + L)^{-1}C'\}$, and a semidefinite program allocates retained information between the identification and precision thresholds. A fixed-grid conditional rate attains the adjudication experiment in the local limit. A fixed-decision release separates adjudication from application in finite samples and measures cost by the number of records read. The application follows Oregon Medicaid lottery invitations through receipt, submission, eligibility decisions, and enrollment. Lottery selection raises documented application receipt by 24.7 percentage points in the full survey frame, with a household-clustered 95 percent interval of 24.1 to 25.4 points. The reporting handoff has loss rank zero for the lottery effect on enrollment, one for population approval following an invitation, and four for approval probabilities across four reception groups; the confidence set for the last rank is $\{4\}$. The fixed-decision release reads 40 to 49 percent of invitation records in expectation and preserves the rank-four threshold. With 25 percent auditable-link capacity, target allocation gives 1.04 times the full-linkage covariance bound for the four-group audit, compared with 1.83 under uniform retention.

1. Introduction

When does a policy become a policy? Economic models date an intervention when a statute, invitation, payment, or interest rate changes. A recipient encounters the intervention after it passes through delivery systems, applications, intermediaries, documents, balance sheets, and attention. Each handoff determines which distinctions reach the recipient's next decision. The records carried across the same handoff determine which counterfactuals an evaluator can recover.

The policy-delivery state carried to recipients differs from the policy-evaluation state carried to later analysts. A handoff may preserve the intervention contrast that changes behavior while discarding the person links needed to recover approval, take-up, or heterogeneous access. The economic object is therefore a sequence of target-specific experiments held by different decision makers along one policy chain.

Policy transmission is represented as a sequence of statistical experiments. At each stage, a parameter-independent Markov kernel transforms the upstream experiment, after which the receiving unit produces a conditional measurement. The handoff removes Fisher information L_j , and the new measurement contributes information A_j . For a vector counterfactual with derivative C , the loss rank ℓ_j counts target directions removed at the handoff. The acquisition rank ρ_j counts those recovered by measurement. Their signed sum gives the change in target ambiguity across the sequence, while weighted covariance bounds record changes in precision at a fixed rank.

In the local Gaussian limit, the upstream score decomposes into the score carried downstream and an independent lost increment. The minimum score rank that restores upstream local identification is

$$\dim C(\ker I_j^-) - \dim C(\ker I_{j-1}).$$

The minimum rank that restores the upstream covariance bound is

$$\text{rank}\{L_j^{1/2} I_{j-1}^{-1} C'\}.$$

The second threshold can lie above the first because a handoff may retain identification while weakening precision. A fixed-grid conditional rate carries the adjudication block in the local experiment. A fixed-decision release reads an iid record stream until m applications appear and releases their approval count; its law is indexed by approval, and its expected record cost is m/a .

For a vector target, a Gaussian retention channel contributes information Q with $0 \leq Q \leq L$. A budget $\text{tr}(WQ) \leq b$ yields a semidefinite program for the weighted covariance bound $\text{tr}\{\Omega C(I + Q)^\dagger C'\}$. Confidence-region inversion produces a set for the loss rank

near singular values, and directional value-function inference covers the repair frontier. Smooth allocations admit the household bootstrap used in the application.

The application uses the Oregon Health Insurance Experiment public-use files (Finkelstein et al. 2012; National Bureau of Economic Research 2026). Oregon drew names from a reservation list in eight lottery draws during 2008. Selection gave household members an opportunity to apply for Medicaid. The initial mail survey records application receipt and submission; administrative files record application, approval, decision date, and subsequent Medicaid enrollment. The arrival analysis follows 13,069 selected survey respondents, the access analysis follows 29,799 selected people in 24,901 households, and the enrollment analysis follows 74,922 people in 66,385 households.

Lottery assignment supplies an experimental check on the survey receipt item. The initial-survey sample contains 58,405 reservation-list members. Documented receipt equals 31.3 percent among selected members and 6.8 percent among controls. A linear probability model with lottery-draw and household-size fixed effects gives a 24.7 percentage-point assignment effect, with a household-clustered 95 percent interval of 24.1 to 25.4 points. Among item respondents, the reported-receipt difference is 56.2 points, and aggregate-monotone response trimming gives an interval from 55.6 to 56.6 points (Lee 2009). Among selected members, reported receipt rises from 15.0 percent during days 0–28 after lottery notification to 89.9 percent during days 56–84. The median interval from notification to an application decision is 76 days.

The reporting exercise follows the records carried across the same program chain. It releases group counts and fixed-grid conditional approval rates, while application information remains at the upstream handoff. The lottery effect on Medicaid enrollment has loss rank zero, population approval following an invitation has rank one, and the four-vector of approval probabilities across requested-language and phone-record groups has rank four. The 95 percent confidence set for the last rank is $\{4\}$. A fixed-decision release set to one quarter of each public group count reads 40 to 49 percent of invitation records in expectation and preserves the rank-four identification threshold. With 25 percent auditable-link capacity, target allocation gives a group-audit risk bound 1.04 times the full-linkage bound; uniform retention gives 1.83. The population-approval target reaches 1.10 times its full-linkage bound at 72 percent capacity.

The comparison-of-experiments literature orders statistical experiments by decision value and information (Blackwell 1953; Goel and DeGroot 1979). Partial sufficiency studies statistics for a subparameter in the presence of nuisance parameters (Basu 1978; Andhivarothai and Ramamoorthi 1994), and missing-information identities quantify score changes under incomplete records (Louis 1982). Rational inattention and information-cost models study chosen experiments and sequential sampling (Miao et al. 2022; Pomatto et al. 2023; Hébert and Woodford 2023). Communication-constrained estimation and

goal-oriented compression ask which summaries carry information for a target (Barnes et al. 2020; Alsing and Wandelt 2018, 2019; Attia et al. 2018). The stage ledger attaches these changes to particular handoffs and measurements and prices repair in score coordinates, records read, or retained person links.

The empirical design also connects to record linkage and administrative access. Statistical agencies jointly choose accuracy, privacy protection, and production resources (Abowd and Schmutte 2019). Record-linkage research studies match quality and the power consequences of link quality (Bailey et al. 2020; Tahamont et al. 2021; Lin and Kolaczyk 2024). Administrative records show how office access, interviews, and procedural rules shape application and enrollment margins (Deshpande and Li 2019; Giannella et al. 2024). The Oregon lottery established the effects of Medicaid access on insurance, utilization, finances, and health (Finkelstein et al. 2012; Baicker et al. 2013). Here the survey and administrative records locate reported arrival, subsequent application, adjudication, and the information carried between those stages.

Section 2 defines the loss-acquisition ledger and arrival clock. Section 3 develops target ranks, risk, and rank confidence sets. Section 4 derives repair channels, record-level releases, the vector frontier, and frontier inference. Section 5 applies the framework to the Oregon lottery. Section 6 concludes.

2. A Policy in Transit

Let the local parameter be $\theta \in \Theta \subseteq \mathbb{R}^p$, and let the policy counterfactual be $\tau : \Theta \rightarrow \mathbb{R}^q$ with derivative $C = D\tau(\theta_0)$. A stage contains a handoff and a measurement. The upstream experiment is $Y_{j-1} \sim P_{\theta,j-1}$. A θ -independent Markov kernel K_j produces the retained record Z_j , and the receiving unit then observes X_j with conditional law $Q_{\theta,j}(\cdot | Z_j)$. The experiment available after the stage is $Y_j = (Z_j, X_j)$.

Write I_{j-1} for upstream Fisher information and I_j^- for the information in Z_j . Data processing gives

$$L_j = I_{j-1} - I_j^- \geq 0.$$

The score of the conditional measurement X_j has conditional mean zero given Z_j , so it is orthogonal to the score of Z_j . If its expected conditional information is $A_j \geq 0$, the stage ends with

$$I_j = I_j^- + A_j.$$

The pair (L_j, A_j) is the information ledger of stage j . The first entry records information removed at the handoff, and the second records information produced by measurement at the receiving unit. A sequence containing only garblings has $A_j = 0$ at every stage. Administrative and financial pipelines commonly contain both entries because an office may receive a reduced record and then collect documents, decisions, or prices of its own.

Define

$$N_{j-1} = \ker I_{j-1}, \quad N_j^- = \ker I_j^-, \quad N_j = \ker I_j = N_j^- \cap \ker A_j$$

and their target images

$$M_{j-1} = C(N_{j-1}), \quad M_j^- = C(N_j^-), \quad M_j = C(N_j).$$

The inclusions $N_{j-1} \subseteq N_j^-$ and $N_j \subseteq N_j^-$ yield two nonnegative ranks:

$$\ell_j = \dim M_j^- - \dim M_{j-1}, \quad \rho_j = \dim M_j^- - \dim M_j.$$

The loss rank ℓ_j counts target directions removed by the handoff. The acquisition rank ρ_j counts target directions recovered by the receiving unit's measurement.

THEOREM 1 (Loss-acquisition ledger). *For every finite sequence of regular stages,*

$$\dim M_J - \dim M_0 = \sum_{j=1}^J (\ell_j - \rho_j).$$

The ranks are invariant to smooth invertible reparameterizations of θ , invertible recodings of each experiment, and invertible linear changes of target coordinates.

The identity attributes final target ambiguity to named handoffs and measurements. A stage may remove several target directions and recover some of them from evidence produced there, giving a net change of $\ell_j - \rho_j$. If every A_j vanishes, the null spaces form a filtration and the identity reduces to the sum of handoff losses.

2.1. Arrival before propagation

Let a nominal intervention occur at date zero, let D_i be the delay before it becomes effective for receiver i , and let g_k be the causal response k periods after effective arrival. If $f_\ell = \mathbb{P}(D_i = \ell)$, the response aligned to the nominal event date is

$$m_h = \sum_{\ell=0}^h f_\ell g_{h-\ell}.$$

PROPOSITION 1 (Arrival convolution). *The nominally aligned response is the convolution $m = f * g$. Over unrestricted real response sequences, a given sequence m admits multiple pairs (f, g) . Knowledge of f with $f_0 > 0$ yields*

$$g_0 = \frac{m_0}{f_0}, \quad g_h = \frac{m_h - \sum_{\ell=1}^h f_\ell g_{h-\ell}}{f_0}.$$

The two sequences describe distinct clocks. The arrival distribution follows delivery, application, notification, account-credit, or processing records. The post-arrival response follows behavior after effective receipt. Linked dates identify the first sequence and permit recursive recovery of the second.

3. Counterfactual Bottlenecks

The loss rank records a change in identification. Fisher information also measures weakening inside the identified region. For a positive-semidefinite information matrix I , the vector target is locally identified exactly when

$$\ker I \subseteq \ker C.$$

Under this range condition its local asymptotic covariance bound is

$$\mathbb{V}(I, C) = CI^\dagger C',$$

where $I^\dagger$ is the Moore–Penrose inverse. Let $\Omega > 0$ specify the units in which components of the target are valued and define

$$\mathcal{R}_\Omega(I, C) = \text{tr}\{\Omega \mathbb{V}(I, C)\}.$$

The value is $+\infty$ when the range condition fails. For a scalar contrast $w'\tau$, the corresponding target strength is $\kappa(I; w)^2 = \{w'\mathbb{V}(I, C)w\}^{-1}$.

The weight matrix has a decision interpretation. Let an institution choose $a \in \mathbb{R}^d$, and suppose local loss around the full-information action $\tau(\theta)$ is $\frac{1}{2}\{a - \tau(\theta)\}'H\{a - \tau(\theta)\}$ with $H > 0$. Using a regular estimator $\hat{\tau}$ as the action gives second-order expected regret $\frac{1}{2}\text{tr}(H\mathbb{V})$. Setting $\Omega = H/2$ makes $\mathcal{R}_\Omega$ the local asymptotic regret bound, while W in Section 4.2 records the resource cost of retaining score information.

At stage j , the risk created by the handoff and the risk removed by measurement are

$$\Delta_{j,\Omega}^{\text{loss}} = \mathcal{R}_\Omega(I_j^-, C) - \mathcal{R}_\Omega(I_{j-1}, C), \quad \Delta_{j,\Omega}^{\text{acq}} = \mathcal{R}_\Omega(I_j^-, C) - \mathcal{R}_\Omega(I_j, C).$$

Both quantities are nonnegative whenever their endpoints are finite. The exact ranks (ℓ_j, ρ_j) locate changes in target identification, while $(\Delta_{j,\Omega}^{\text{loss}}, \Delta_{j,\Omega}^{\text{acq}})$ record precision changes that preserve rank. This pairing separates a handoff that deletes a target direction from a handoff that leaves the direction weakly measured.

COROLLARY 1 (Reception regret). *Under the local quadratic decision problem with $\Omega = H/2$, $\Delta_{j,\Omega}^{\text{loss}}$ is the increase in the receiver's local asymptotic regret bound created by the handoff, and $\Delta_{j,\Omega}^{\text{acq}}$ is the decrease produced by measurement at the receiving stage.*

The corollary attaches each experiment to a decision maker. A household may decide whether to submit an application after receiving it. An agency may issue an eligibility decision after reviewing documents, and an evaluator may estimate a target after a reporting handoff. The matrix H records the loss from errors in the relevant decision, while the stage ledger measures the resulting change in attainable regret.

3.1. Rank estimation under separated spectra

Let P_{j-1} and P_j^- denote the orthogonal projections onto N_{j-1} and N_j^- . For a matrix B , write $\text{rank}_c(B)$ for the number of singular values above c . With estimators $\widehat{I}_{j-1}$, $\widehat{I}_j^-$, and $\widehat{C}$, construct the corresponding small-eigenvalue projections $\widehat{P}_{j-1}(c)$ and $\widehat{P}_j^-(c)$ and set

$$\widehat{\ell}_j(c) = \text{rank}_c\{\widehat{C}\widehat{P}_j^-(c)\} - \text{rank}_c\{\widehat{C}\widehat{P}_{j-1}(c)\}.$$

PROPOSITION 2 (Separated-rank estimation). *Allow the parameter and target dimensions to vary with n . Suppose $\|\widehat{I}_{j-1} - I_{j-1}\| + \|\widehat{I}_j^- - I_j^-\| + \|\widehat{C} - C\| = O_p(a_n)$, $\|C\|$ is bounded, and the positive eigenvalues that define the two null spaces and the positive singular values of CP_{j-1} and CP_j^- exceed η_n . If*

$$\frac{a_n}{\eta_n} = o(c_n), \quad c_n = o(\eta_n),$$

then $\widehat{\ell}_j(c_n) \rightarrow_p \ell_j$.

The proposition describes point estimation at a parameter with separated ranks. Its rate conditions cover a growing target list when operator-norm estimation error remains below the spectral separation scale. Projection of the estimated information pair onto $\widehat{I}_{j-1} \geq \widehat{I}_j^- \geq 0$ enforces the population ordering before the calculation. The Oregon target list contains four pre-specified reception groups, so its reported rank set uses the finite-dimensional confidence-region construction below.

3.2. Confidence sets near rank changes

Let $\beta_j = (I_{j-1}, I_j^-, C)$ and let $\mathfrak{P}_j(r)$ collect primitive triples that satisfy the information ordering and have loss rank r . Suppose $\mathfrak{C}_{n,j}(1 - \alpha)$ is a joint confidence region for β_j with coverage uniform over a class $\mathfrak{P}_j$ that permits eigenvalues and target singular values of order $n^{-1/2}$. Define

$$\mathfrak{R}_{n,j}(1 - \alpha) = \{r \in \{0, \dots, q\} : \mathfrak{C}_{n,j}(1 - \alpha) \cap \mathfrak{P}_j(r) \neq \emptyset\}.$$

PROPOSITION 3 (Weak-rank confidence set). *If $\mathfrak{C}_{n,j}(1-\alpha)$ has uniform asymptotic coverage, then*

$$\liminf_{n \rightarrow \infty} \inf_{\beta_j \in \mathfrak{P}_j} \mathbb{P}_{\beta_j} \{\ell_j \in \mathfrak{R}_{n,j}(1-\alpha)\} \geq 1 - \alpha.$$

At separated primitives, the set converges in probability to $\{\ell_j\}$.

The construction inverts a confidence region for the objects that generate rank, so a local singular value remains in the reported set. A Wald ellipsoid or a bootstrap confidence region for the vectorized score moments supplies $\mathfrak{C}_{n,j}$. In moderate dimensions, computation enumerates r and minimizes distance to $\mathfrak{P}_j(r)$. When the null-space bases are separated, the induced quotient map can be tested directly with the sum of its trailing squared singular values; bootstrap rank tests developed by Chen and Fang (2019) provide critical values for that calculation. The Oregon application has a diagonal quotient map, and inversion of simultaneous confidence intervals gives the rank set in closed form.

4. The Architecture of Reception

4.1. A channel drawn from the upstream experiment

Fix a handoff and write $I = I_j^-$ and $L = L_j$, so the upstream information is $I + L = I_{j-1}$. Its local Gaussian limit admits independent central sequences with information I and L . Write the lost increment as

$$U \sim \mathcal{N}(L^{1/2}h, \mathbf{I}_p)$$

under local parameter h . A regular k -coordinate retention channel has Gaussian limit $T_R = R'U$, where R has k columns. Its information is

$$Q_R = L^{1/2}P_R L^{1/2},$$

with P_R the orthogonal projection onto the column space of R . This definition ties the channel to data held upstream: T_R is a statistic of the lost score increment generated by a parameter-independent Markov kernel.

THEOREM 2 (Minimum identification repair). *Among regular Gaussian retention channels, the minimum score rank required to preserve the upstream local identification of the target is*

$$k^* = \dim C(\ker I) - \dim C\{\ker(I + L)\} = \ell_j.$$

There is a k^ -coordinate channel satisfying*

$$C\{\ker(I + Q_R)\} = C\{\ker(I + L)\},$$

and every channel satisfying this equality has score rank at least k^* .

The theorem concerns a statistic constructed from the upstream experiment and preserves the target's local identification set. Matching precision within that set imposes an additional requirement. The proof works on $\ker I / \ker(I + L)$. The quadratic form induced by L is an inner product on this quotient, so every target functional there can be represented as a linear functional of $L^{1/2}h$. A basis of the target image gives ℓ_j retained score coordinates. Score rank measures local statistical dimension; a database may encode the resulting statistic in several fields or in one composite record.

Full-information precision generally requires a larger channel. Let $F = I + L$ and suppose $F > 0$. Define

$$k^{\text{prec}} = \text{rank}\{L^{1/2}F^{-1}C'\}.$$

THEOREM 3 (Minimum precision repair). *Among Gaussian retention channels, the minimum score rank that attains the upstream covariance bound*

$$C(I + Q_R)^\dagger C' = CF^{-1}C'$$

is k^{prec} . An attaining channel has information

$$Q^{\text{prec}} = L^{1/2}P_GL^{1/2}, \quad G = L^{1/2}F^{-1}C',$$

where P_G projects onto the column space of G . Moreover,

$$k^{\text{prec}} \geq \ell_j.$$

The matrix G collects the lost-score components used by the full-information efficient influence function for the target. Retaining their span reproduces the upstream covariance bound. The inequality separates two design thresholds. The first restores local identification; the second restores full-information precision. The repair frontier below traces the interval between them.

PROPOSITION 4 (Pilot implementation). *Suppose the upstream score and the repair loading $R(\theta)$ are continuous in quadratic mean near θ_0 . Let an independent pilot produce $\tilde{\theta} \rightarrow_p \theta_0$, and apply the channel with loading $R(\tilde{\theta})$ to the production records. Its conditional information converges in probability to $Q_{R(\theta_0)}$. Under spectral separation, the estimated channel has the same score rank with probability approaching one, and its target ambiguity and covariance converge to the corresponding repair benchmark.*

The pilot and production records together define a parameter-independent procedure. The local channel can therefore be implemented from an estimated model. Designs based

directly on record sampling, including Proposition 7, implement their information menu through the sampling rule itself.

PROPOSITION 5 (Conditional-rate release). *Let $(A_i, A_i Y_i)_{i=1}^n$ be independent with $\mathbb{P}(A_i = 1) = a$ and $\mathbb{P}(Y_i = 1 \mid A_i = 1) = b$, where (a, b) lies in a compact subset of $(0, 1)^2$. Set $K_n = \sum_i A_i$, $R_n = \sum_i A_i Y_i$, and $\widehat{b}_n = R_n/K_n$, using any fixed value on $K_n = 0$. For a public grid width δ_n satisfying $\sqrt{n}\delta_n \rightarrow 0$ and $n\delta_n \rightarrow \infty$, release $\widetilde{b}_n = \delta_n \text{round}(\widehat{b}_n/\delta_n)$. Under $(a_n, b_n) = (a_0 + h_a/\sqrt{n}, b_0 + h_b/\sqrt{n})$,*

$$\sqrt{n}(\widehat{b}_n - b_0) \rightsquigarrow \mathcal{N}\left(h_b, \frac{b_0(1-b_0)}{a_0}\right).$$

The release has local information $\text{diag}\{0, a_0/[b_0(1-b_0)]\}$ for (a, b) . It carries the adjudication-score information of the linked experiment and carries zero root- n information about the application probability.

The rounded conditional rate is a parameter-independent statistic of the production records. Its grid has a public denominator independent of applicant count. Its fixed output dimension attains the adjudication information in the local experiment. Group-specific conditional rates apply the construction componentwise. Partial-sufficiency theory studies statistics for a subparameter in the presence of nuisance parameters (Basu 1978; Andhivarothai and Ramamoorthi 1994). The reception design begins with a named downstream experiment and asks which upstream score coordinates restore a counterfactual derivative at each identification and precision threshold.

The local release attains the adjudication block because its precision scales with the realized number of applications. Exact finite-sample separation fixes the number of adjudication outcomes and places the random record-reading cost upstream. The stopping rule is inverse binomial sampling (Feller 1968). Let $\tau_r = \inf\{s : \sum_{i=1}^s A_i = r\}$ be the location of the r th application in an iid record stream.

PROPOSITION 6 (Exact separation and record cost). *Fix $m \geq 1$ and release $T_m = \sum_{r=1}^m Y_{\tau_r}$ while retaining $(\tau_1, \dots, \tau_m)$ upstream. Then*

$$T_m \sim \text{Binomial}(m, b),$$

so the release has Fisher information $\text{diag}\{0, m/[b(1-b)]\}$ for (a, b) and $\mathbb{E}[\tau_m] = m/a$. Any release whose distribution is invariant to a over an interval has b information invariant to a over that interval. The full adjudication information from n invitations is $na/[b(1-b)]$, so coincidence with full adjudication precision occurs for at most one value of a within that interval.

The proposition describes two production choices. A fixed-grid rate attains full adjudication precision in the local experiment. A fixed-decision release separates adjudication from application in finite samples and prices production by the number of records read.

Both carry a positive b score and preserve the same local identification directions, although their precision and record costs differ.

COROLLARY 2 (Placement). *Every statistic formed solely from the retained experiment has a null space containing $\ker I$ and target ambiguity containing $C(\ker I)$. Preservation of the directions in $C(\ker I)/C\{\ker(I + L)\}$ therefore occurs at the handoff or upstream. Evidence collected after the handoff enters the ledger as acquisition information A_j .*

The placement statement distinguishes two operations with the same eventual identification outcome. Retention carries an available distinction across a handoff. Acquisition creates a measurement at the receiving stage. Their costs, legal authorities, and data requirements can differ even when $I + Q$ and $I + A$ have the same numerical value.

4.2. The information menu and vector repair frontier

Gaussian garblings of the lost increment generate the information menu

$$\mathcal{Q}(L) = \{Q : 0 \leq Q \leq L\}.$$

Every element can be written $Q = L^{1/2}PL^{1/2}$ for a positive-semidefinite contraction P . Its score rank is $\text{rank } Q$, and a Gaussian channel with that score rank implements it. This menu is the local form of data processing; a proposed record-retention rule is feasible when its score information belongs to $\mathcal{Q}(L)$. In an applied design, the map from records to Q supplies the units of the budget. Section 5 uses independent stratified retention of person-level links, so q_g is an expected record share and the budget is the expected fraction of links retained.

PROPOSITION 7 (Stratified link retention). *Suppose the lost score is a sum of independent person-level contributions from groups $g = 1, \dots, G$, with group share p_g and conditional information L_g . Retain each upstream–downstream link in group g independently with probability q_g , and record the sampling indicator. The retained links contribute information*

$$Q(q) = \sum_{g=1}^G p_g q_g L_g$$

per person and use expected record capacity $\sum_g p_g q_g$. Thus a capacity bound B gives the feasible set $\sum_g p_g q_g \leq B$, with $0 \leq q_g \leq 1$.

The sampling rule uses record membership and an external random draw, so production operates directly on record strata. Group weights in the target determine the allocation across q_g , while p_g converts each allocation into an expected number of retained links.

Let $W > 0$ price retained information, let b be a budget, and let $\Omega > 0$ weight a q -dimensional counterfactual. Define the repair frontier

$$\Phi(b; I, L, W, C, \Omega) = \min_{Q, V} \text{tr}(\Omega V)$$

subject to

$$\begin{pmatrix} I + Q & C' \\ C & V \end{pmatrix} \succeq 0, \quad 0 \leq Q \leq L, \quad \text{tr}(WQ) \leq b.$$

The value is $+\infty$ when finite V is infeasible. The block constraint then records failure of target identification directly.

THEOREM 4 (Vector repair frontier). *The displayed program equals*

$$\min_{0 \leq Q \leq L, \text{tr}(WQ) \leq b} \mathcal{R}_\Omega(I + Q, C).$$

Its value is decreasing and convex in b . Full retention gives $\Phi\{\text{tr}(WL)\} = \mathcal{R}_\Omega(I + L, C)$. Suppose an optimizer Q_b satisfies $I + Q_b > 0$ and Slater's condition. With

$$H_b = (I + Q_b)^{-1} C' \Omega C (I + Q_b)^{-1},$$

there are $A_b, S_b \geq 0$ and $\lambda_b \geq 0$ such that

$$-H_b + \lambda_b W - A_b + S_b = 0,$$

$$\text{tr}(A_b Q_b) = 0, \quad \text{tr}\{S_b(L - Q_b)\} = 0, \quad \lambda_b\{\text{tr}(WQ_b) - b\} = 0.$$

At differentiability points,

$$\partial_b \Phi = -\lambda_b, \quad D_L \Phi[D] = -\text{tr}(S_b D)$$

for a feasible positive-semidefinite menu expansion D .

The scalar case $q = 1$ yields the c -optimal design criterion (Elfving 1952; Sagnol 2011). A vector target can allocate score rank across several counterfactual coordinates. The matrix S_b records which unavailable score directions constrain the target, while λ_b gives the reduction in weighted target variance produced by one unit of retention budget.

With orthogonal lost increments across handoffs, the institution can solve one program with $Q_j \leq L_j$ and $\sum_j \text{tr}(W_j Q_j) \leq B$. The stage index on each cap assigns every retained score to the final location at which the source record exists. Sequential information acquisition and rational-inattention models price experiments or posterior beliefs (Miao et al. 2022;

Pomatto et al. 2023; Hébert and Woodford 2023); the caps here attach those choices to the score increments removed at specific handoffs.

4.3. Inference for the frontier

Let $\beta = (I, L, W, C, \Omega)$ and $\phi_b(\beta) = \Phi(b; \beta)$. Suppose $\sqrt{n}(\widehat{\beta} - \beta) \rightsquigarrow Z$ and a cluster or block bootstrap consistently estimates the law of Z . Suppose also that, on the compact budget set used for inference, target identification holds with a uniform margin and the parametric semidefinite value map is continuous and Hadamard directionally differentiable uniformly over the reported finite budget grid. Then $\sqrt{n}\{\phi_b(\widehat{\beta}) - \phi_b(\beta)\}$ converges to the directional derivative $\phi'_{b,\beta}(Z)$. Active caps can give this derivative a nonlinear form.

For a bootstrap draw $\widehat{\beta}^*$, let $Z_n^* = \sqrt{n}(\widehat{\beta}^* - \widehat{\beta})$ and choose $\varepsilon_n \downarrow 0$ with $\sqrt{n}\varepsilon_n \rightarrow \infty$. The numerical delta draw is

$$D_{n,b}^* = \frac{\phi_b(\widehat{\beta} + \varepsilon_n Z_n^*) - \phi_b(\widehat{\beta})}{\varepsilon_n}.$$

PROPOSITION 8 (Frontier bands). *Under the preceding conditions, the conditional law of $D_{n,b}^*$ consistently estimates the law of $\phi'_{b,\beta}(Z)$ at each budget. On a finite budget grid, quantiles of the maximum centered numerical-delta process give simultaneous bands. If the weak-rank confidence region includes primitives that violate target identification at a budget, the corresponding upper endpoint is $+\infty$.*

The numerical delta construction follows the resampling logic for directionally differentiable maps in Fang and Santos (2019) and Hong and Li (2018). Value-function inference supplies related uniform procedures (Firpo et al. 2023). In the Oregon specialization, positive group coefficients and a scalar capacity constraint yield a unique allocation. The analysis uses the ordinary household bootstrap and a simultaneous max- t band over the displayed capacity grid. The Online Appendix reports ordinary-bootstrap and numerical-delta coverage in a design calibrated to the four reception groups.

5. Arrival and Reception in the Oregon Medicaid Lottery

5.1. Invitation and reported arrival

Oregon opened a reservation list for the Oregon Health Plan Standard program in 2008 and selected names by lottery in eight draws. Selection notified household members that they could submit an application; eligibility review then determined approval. The lottery design and subsequent outcomes are described by Finkelstein et al. (2012). The public-use files link the reservation-list record, lottery draw, notification date, initial mail survey, application indicator, application-decision date, approval indicator, and

subsequent Medicaid enrollment under person and household identifiers (National Bureau of Economic Research 2026).

The initial mail survey asks, “As of today, have you received an application?” Its sampling frame contains 58,405 reservation-list members in 50,094 households, including lottery-selected members and controls.

TABLE 1. Lottery selection and reported application receipt

Outcome	Control	Selected	Adjusted difference
Receipt item response	44.6	44.2	-0.4 [-1.3, 0.5]
Documented receipt	6.8	31.3	24.7 [24.1, 25.4]
Reported receipt	15.1	70.9	56.2 [55.2, 57.2]
Lee trimming interval	trim 1.04% of controls		55.6–56.6

Notes. The initial survey sample contains lottery-selected and control reservation-list members. Means are percentages. Adjusted differences are coefficients on lottery selection in linear probability models with lottery-draw and household-size fixed effects and household-clustered 95% intervals. Documented receipt equals one when a sampled member answers the item and reports receipt, so its assignment contrast uses the full survey frame. Reported receipt conditions on item response. The trimming interval imposes aggregate monotone response selection and matches the lower item-response rate by trimming the assignment group with the higher rate.

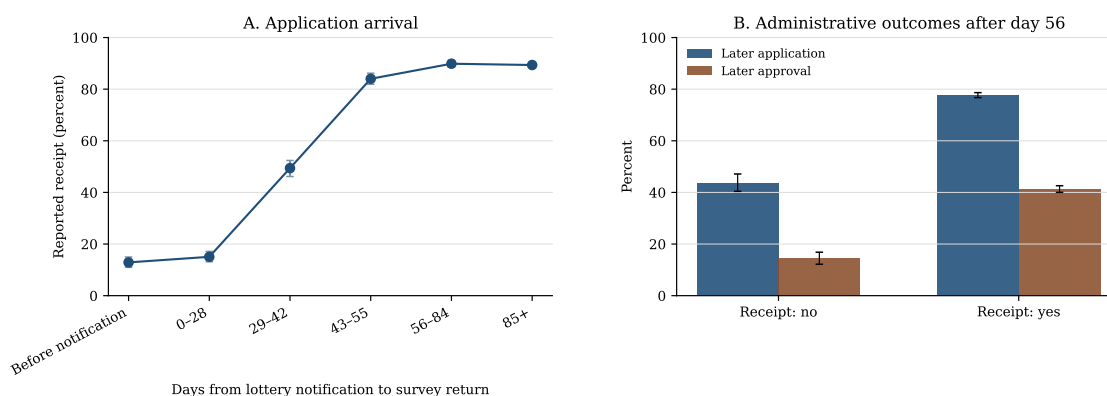
Table 1 uses lottery assignment to validate the receipt item. Item-response rates are 44.2 percent among selected members and 44.6 percent among controls, with an adjusted difference of –0.4 percentage point and a 95 percent interval from –1.3 to 0.5. Define documented receipt as an item response that reports receipt. Its mean is 31.3 percent among selected members and 6.8 percent among controls. A linear probability model with lottery-draw and household-size fixed effects gives a 24.7 point assignment effect, with a household-clustered interval from 24.1 to 25.4. This outcome is defined for all 58,405 sampled members. Among item respondents, reported receipt is 70.9 percent for selected members and 15.1 percent for controls, with an adjusted difference of 56.2 points and an interval from 55.2 to 57.2. Under aggregate monotone response selection, trimming the 1.04 percent excess response share among controls gives a 55.6 to 56.6 point interval (Lee 2009). The full-frame documented-receipt effect supplies the assignment-based validation; the respondent contrast and trimming interval describe its receipt component.

The arrival analysis then conditions on lottery selection and links 13,069 responses with observed survey-return and notification dates. Figure 1 treats the survey answer as a current-status measure of arrival. Reported receipt is 15.0 percent during days 0–28 after notification, 49.4 percent during days 29–42, 84.0 percent during days 43–55, and 89.9 percent during days 56–84. The 12.9 percent report before the focal notification date

records household timing and reporting error contained in this measure. Delivery time remains interval-censored between dated observations.

Among survey responses after day 55 with complete administrative outcomes, 7,383 people report receipt and 867 report nonreceipt. Later application rates are 77.7 and 43.7 percent, and later approval rates are 41.4 and 14.4 percent. Survey response timing and reported receipt are selected outcomes, so the panel describes continuation through the administrative chain. Among 2,569 selected respondents who had yet to submit and answered the reason question, 40.4 percent report unfinished work, 8.8 percent report paperwork hassle, 5.7 percent report missing citizenship documents, and 6.5 percent report missing other documents; the survey permits several reasons.

FIGURE 1. Reported arrival and subsequent administrative outcomes



Notes. Panel A reports application receipt as of the initial mail survey among 13,069 lottery-selected respondents with observed survey-return and notification dates. Panel B reports later administrative application and approval among respondents observed after day 55, separated by reported receipt. Bars give household-bootstrap 95% percentile intervals from resampling households within lottery draw. The comparisons describe selected survey respondents.

The administrative decision clock is observed for 18,100 selected applicants with a decision date on or after notification. Its twenty-fifth, fiftieth, seventy-fifth, and ninetieth percentiles are 55, 76, 98, and 120 days. An event study aligned to lottery notification combines arrival, submission, review, and post-decision response. Proposition 1 separates the arrival distribution from the response after effective receipt when linked timing data identify the former.

5.2. A linked two-step experiment

The preceding evidence concerns the household's encounter with the mailed application. The next exercise concerns the information available to an evaluator after an agency

reporting handoff. Both experiments follow the same program chain and attach their information sets to separate decisions.

The access analysis continues to condition on lottery selection because its target is reception of an issued invitation. Of 29,834 selected people, 35 have an incomplete application or approval field, leaving 29,799 people in 24,901 households. The invitation records define four groups from the language requested for mailed materials and the presence of a phone number at signup. These fields precede lottery selection. The lottery selected with probabilities that varied by the number of household members on the list; inverse selection weights enter a robustness calculation.

For group $g \in \{1, \dots, 4\}$, let A_i indicate application after invitation and let Y_i indicate approval for an applicant. Write

$$a_g = \mathbb{P}(A_i = 1 \mid g, \text{selected}), \quad b_g = \mathbb{P}(Y_i = 1 \mid A_i = 1, g, \text{selected}), \quad d_g = a_g b_g.$$

The target is $d = (d_1, \dots, d_4)'$, the vector of approval probabilities following an invitation. Conditional on group counts n_g , the linked likelihood factorizes into the application likelihood and the approval likelihood among applicants. Their scores are orthogonal, and their information matrices are diagonal across groups with entries

$$I_{a,g} = \frac{n_g}{a_g(1 - a_g)}, \quad I_{b,g} = \frac{n_g a_g}{b_g(1 - b_g)}.$$

For each group, let $K_g = \sum_{i:g_i=g} A_i$, $R_g = \sum_{i:g_i=g} A_i Y_i$, and $\widehat{b}_g = R_g/K_g$. The evaluated reporting handoff releases the invitation-frame count n_g and fixed-grid rate $\widetilde{b}_g = \delta_g \text{round}(\widehat{b}_g/\delta_g)$ together with the lottery-enrollment experiment. Applicant totals and invitation-application person links remain upstream. The application sets $\delta_g = n_g^{-3/4}$, a public resolution independent of K_g . Proposition 5 establishes that $\widetilde{b}_g$ carries information $n_g a_g / [b_g(1 - b_g)]$ for b_g and zero root- n information for a_g . The statistic is a parameter-independent function of the production records, so the handoff implements the retained adjudication experiment in the local limit. At finite n_g , its sampling distribution depends on a_g through the random denominator; this dependence enters below the root- n_g information scale used by the theory. Across the four Oregon groups, rounding changes the conditional approval rates by at most 0.6 percentage point. The public-use data contain the linked source experiment and permit evaluation of the release. Since $\partial d_g / \partial a_g = b_g$, the quotient target map for the lost increment is

$$\text{diag}(b_1, b_2, b_3, b_4).$$

Its rank gives the number of invitation-application score coordinates required to preserve the four approval probabilities.

Proposition 6 supplies a finite-sample release whose law is invariant to a_g . Place invitation records in an external random order, read them until m_g applications appear, release the approval count among those m_g decisions, and retain the stopping location upstream. The Oregon design sets $m_g = \lfloor n_g/4 \rfloor$ from the public invitation-frame count.

TABLE 2. Exact separation through a fixed-decision release

Group	Decisions	Records read	Approval rate
English requested, phone	5,804	39.7%	47.9%
English requested, phone absent	915	48.7%	51.5%
Other language requested, phone	663	39.4%	35.9%
Other language requested, phone absent	67	45.1%	50.7%

Notes. The release places invitation records in an external random order, reads records until one quarter of each group count has applied, and reports the approval count. Records read are realized percentages of group invitation counts. Expected reading shares range from 40.0 to 49.0 percent and equal the shares of linked adjudication information retained.

The fixed-decision release reads 40.0 to 49.0 percent of invitation records in expectation and carries the same share of each group’s full adjudication information. Its four positive binomial score coordinates preserve the group target’s rank-four identification threshold. The fixed-grid release supplies the full-precision local benchmark, while the fixed-decision release supplies exact separation with an observed record-reading unit.

TABLE 3. Target-specific ranks at the reporting handoff

Target	Estimate	Loss rank	Repair ranks
Lottery effect on Medicaid enrollment	25.6 [24.9, 26.3]	0	0 / 0
Approval following invitation, population	29.2 [28.6, 29.8]	1	1 / 1
Approval following invitation, four groups	23.3–30.4	4 {4}	4 / 4

Notes. The reporting handoff releases group conditional approval rates and the lottery–enrollment experiment. Applicant totals and invitation–application links remain upstream. The first estimate is the coefficient on lottery selection in a regression with lottery-draw and household-size fixed effects; its interval clusters by household. The other intervals resample households within draw. Repair ranks report minimum identification rank followed by minimum full-precision rank. The braces give the 95% confidence set for the four-group loss rank. Estimates are percentage points in the first row and percentages in the remaining rows.

Table 3 evaluates the same reporting handoff under three targets. The lottery effect on subsequent Medicaid enrollment uses the retained lottery–enrollment link and has loss rank zero. A regression with lottery-draw and household-size fixed effects estimates an enrollment effect of 25.6 percentage points, with a household-clustered 95 percent interval

of 24.9 to 26.3 points. The population approval probability uses one application-score contrast and has loss rank one. The four group approval probabilities use four contrasts and have loss rank four. The handoff therefore has a target-specific effect within one experiment.

5.3. Where reception changes

TABLE 4. Reception of the Oregon Medicaid invitation

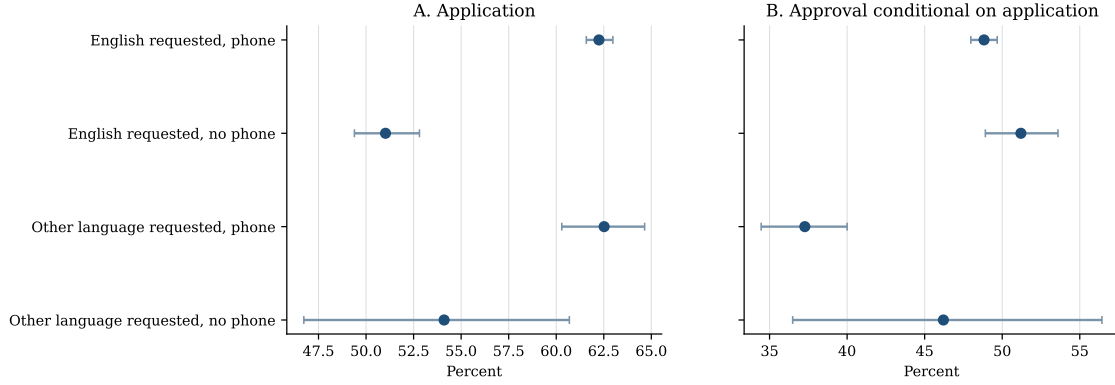
Lottery-list record	People	Applied	Approved applied	Approved
English requested, phone	23,216	62.2	48.8	30.4
English requested, phone absent	3,663	51.0	51.2	26.1
Other language requested, phone	2,652	62.5	37.3	23.3
Other language requested, phone absent	268	54.1	46.2	25.0

Notes. Rates are percentages among lottery-selected people with complete administrative application and approval fields. The third column conditions on a submitted application. Approval is the application decision recorded by Oregon. Confidence intervals and tests resample households within lottery draw.

The invitation produces an application for 60.8 percent of selected people. Approval conditional on application is 48.0 percent, so 29.2 percent of selected people receive approval. Table 4 and Figure 2 separate these margins by the two contact fields.

Within records requesting English, the application rate is 62.2 percent when a phone number is recorded and 51.0 percent when the phone field is empty, a difference of –11.2 percentage points. Conditional approval differs by 2.4 points across those records. Among records with a phone number, application rates differ by 0.3 points across language requests, while conditional approval is 11.6 points lower for records requesting another language. The Online Appendix reports household-bootstrap intervals for these contrasts. The location of the gap therefore changes across the two record comparisons: phone availability is associated with the application margin, and language request is associated with the adjudication margin among applicants.

FIGURE 2. Application and conditional approval after lottery selection



Notes. Points are group rates among 29,799 selected people with complete application and approval records. Bars are household-bootstrap 95% percentile intervals from resampling households within lottery draw. Language request and phone availability come from the reservation-list record.

5.4. Loss rank and minimum repair

TABLE 5. Information at the invitation–application reporting handoff

Group	Application	Decision	b_g	95% interval
English requested, phone	98,790	57,836	0.488	[0.478, 0.499]
English requested, phone absent	14,658	7,480	0.512	[0.483, 0.541]
Other language requested, phone	11,317	7,091	0.373	[0.338, 0.407]
Other language requested, phone absent	1,079	583	0.462	[0.337, 0.587]
Target-loss rank	4; simultaneous 95% confidence set {4}			

Notes. Application information is $n_g/[a_g(1 - a_g)]$. Decision information is $n_g a_g/[b_g(1 - b_g)]$. The linked source contains both scores. The release reports a rounded conditional rate on a public grid with width $n_g^{-3/4}$ and has the displayed decision information in its local limit. Raw and released conditional rates differ by at most 0.6 percentage point. The approval target is $d_g = a_g b_g$; the missing-score derivative is b_g . The simultaneous intervals use the household bootstrap and establish four target directions.

Every conditional approval estimate lies between 0.373 and 0.512. Simultaneous household-bootstrap intervals for all four b_g coefficients exclude zero. Inverting their joint confidence box gives a 95 percent loss-rank confidence set of {4}. Theorem 2 then gives four score coordinates for the group approval vector. One coordinate per group can carry the local application score through the reporting handoff.

The repair dimension changes with the target. For the population approval probability $\bar{d} = \sum_g p_g d_g$, the derivative on the lost coordinates is $(p_1 b_1, \dots, p_4 b_4)$, which has rank one. A single weighted application-score coordinate restores its local identification. For

the group vector, four coordinates restore local identification. Theorem 3 gives the same ranks for full-information precision in this orthogonal binomial model. These ranks describe score dimension. The baseline release carries four conditional rates. A producer authorized to compute target-specific summaries can add four application totals to attain the group-target rank bound. Proposition 7 supplies a record-level implementation for a release that preserves auditable person links for subsequent analyses.

5.5. The estimated repair frontier

Retain each invitation–application link in group g independently with probability q_g . Proposition 7 gives application information $q_g I_{a,g}$ and expected person-link capacity $\sum_g p_g q_g$. With equal weight on the four approval probabilities, the large-sample risk multiplied by N is

$$\mathcal{V}(q) = \sum_{g=1}^4 \left\{ \frac{b_g^2 a_g (1 - a_g)}{p_g q_g} + \frac{a_g b_g (1 - b_g)}{p_g} \right\}.$$

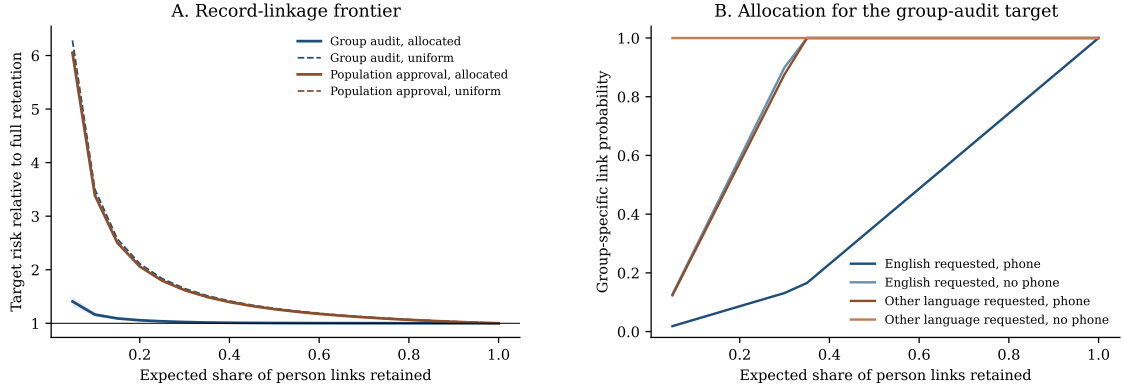
The first term prices the application score carried across the handoff; the second comes from the approval score retained in the adjudication file. The record menu consists of sampled person links, with application information determined by those retained links. For record-share capacity $\sum_g p_g q_g \leq B$, the solution is

$$q_g(B) = \min \left\{ 1, \frac{b_g \{a_g (1 - a_g)\}^{1/2}}{p_g \sqrt{\lambda(B)}} \right\},$$

where $\lambda(B)$ clears the budget. This is the diagonal specialization of Theorem 4.

Figure 3 reports two target frontiers and the group-audit allocation under this auditable-link menu. At 25 percent link capacity, allocation by target risk gives 1.04 times the full-linkage bound for the group audit, while uniform retention gives 1.83. The allocated link probabilities are 0.109, 0.746, 0.726, and 1.000 in the order shown in Table 4. The smallest group reaches full linkage because each group probability receives the same target weight and its records use less capacity. Fourteen percent link capacity reaches 1.10 times the full-linkage group-audit bound. For the population approval probability, group weights enter the target and 72 percent capacity is required to reach the same ratio. The score-summary channel and the auditable-link menu answer separate release questions; the first minimizes local statistical dimension, while the second prices retained microdata records.

FIGURE 3. Repair frontier for the four approval probabilities



Notes. Panel A reports covariance bounds relative to full retention. Solid lines allocate link probabilities by target risk; dashed lines retain a common fraction in every group. The shaded region is a simultaneous 95% household-bootstrap band for the group-audit frontier. Panel B reports group-specific link probabilities for the group-audit target. A capacity of one permits full linkage.

TABLE 6. Precision under a record-linkage capacity constraint

Linked record share	Audit, allocated	Audit, uniform	Population, allocated	Population, uniform
10%	1.16	3.50	3.39	3.40
25%	1.04	1.83	1.80	1.80
50%	1.01	1.28	1.26	1.27

Notes. Entries are covariance bounds divided by the corresponding full-linkage bound. Audit gives equal weight to the four group approval probabilities. Population uses the approval probability for the selected population. Allocated retention solves the record-share frontier; uniform retention links the displayed share in every group.

5.6. Sampling and interpretation

The bootstrap resamples households within each of the eight lottery draws and carries variation in application rates, conditional approval rates, and group shares through the frontier. Positive coefficients and the capacity constraint give a unique Oregon allocation, so the household bootstrap supplies a simultaneous max- t band over the displayed capacity grid. The rank set comes from a simultaneous max- t interval for $(b_1, \dots, b_4)$. The Online Appendix reports approval rates after inverse lottery-selection weighting and after deletion of each draw. Weighted rates differ from the selected-cohort rates by at most 1.5 percentage points. Across draw deletions, group approval rates stay within 29.3–31.6, 25.2–27.3, 22.1–24.0, and 22.0–27.1 percent.

The contact-field comparisons describe reception groups; the lottery randomizes invitation opportunity conditional on household size and leaves the signup fields as observed characteristics. The information calculation evaluates a reporting architecture using linked experimental records. Its conclusion concerns retention of the invitation-application link for measuring approval following an invitation. Effects of changing phone access, language services, or eligibility rules require interventions on those components. Record-linkage studies quantify errors and power losses created when experimental and administrative files are matched (Bailey et al. 2020; Tahamont et al. 2021). The target comparison holds the linked source fixed and asks which score components a release must carry for each estimand.

6. Conclusion

A policy reaches an economic decision through handoffs and measurements. The information ledger assigns target loss to a handoff and target acquisition to evidence produced at the receiving stage. Its exact ranks record changes in counterfactual identification, and its covariance increments record changes in precision.

The minimum identification-repair rank equals the target-loss rank. The minimum rank that restores the upstream covariance bound is the rank of the lost component of the target's full-information influence function. A vector repair frontier allocates retained information between these thresholds under stage-specific availability and cost. Confidence-region inversion carries rank uncertainty through weak singular values, and value-function methods carry sampling variation through the estimated frontier.

The Oregon lottery records show how these objects operate in a linked program-access chain. Lottery selection raises documented application receipt by 24.7 percentage points in the full initial-survey frame, with an interval from 24.1 to 25.4 points. Reported receipt among selected members rises from 15.0 percent during days 0–28 after notification to 89.9 percent during days 56–84. The median interval from notification to an application decision is 76 days. A release of fixed-grid group conditional approval rates then has loss rank zero for the lottery effect on Medicaid enrollment, one for population approval following an invitation, and four for the group approval vector. Proposition 5 supplies the full-precision local release. Proposition 6 supplies the exact separated release; at one quarter of each public group count, it reads 40 to 49 percent of invitation records in expectation and preserves the rank-four threshold. Target-specific application summaries attain the repair bounds. Under an auditable-link menu, stratified random retention converts the frontier into expected person-record capacity, and the allocation changes across a population target and an equal-weight group audit.

The same accounting applies whenever upstream and downstream score records can be linked, including payment delivery, benefit enrollment, bank underwriting, tax filing,

procurement, and monetary transmission. The empirical design begins with a target. It then records the experiment held on each side of a handoff and assigns each information set to its decision maker. Recipient records locate arrival and subsequent action, while reporting records determine which counterfactuals an evaluator can recover. Together these records show when an announced action becomes an intervention that both the economy and its evidence can carry forward.

References

- Abowd, John M. and Ian M. Schmutte**, “An Economic Analysis of Privacy Protection and Statistical Accuracy as Social Choices,” *American Economic Review*, 2019, 109 (1), 171–202.
- Alsing, Justin and Benjamin Wandelt**, “Generalized Massive Optimal Data Compression,” *Monthly Notices of the Royal Astronomical Society: Letters*, 2018, 476 (1), L60–L64.
- **and —**, “Nuisance Hardened Data Compression for Fast Likelihood-Free Inference,” *Monthly Notices of the Royal Astronomical Society*, 2019, 488 (4), 5093–5103.
- Andhivarthai, N. and R. V. Ramamoorthi**, “Partial Sufficiency and Least Favorable Experiments,” *Scandinavian Journal of Statistics*, 1994, 21 (4), 493–500.
- Attia, Ahmed, Alen Alexanderian, and Arvind K. Saibaba**, “Goal-Oriented Optimal Design of Experiments for Large-Scale Bayesian Linear Inverse Problems,” *Inverse Problems*, 2018, 34 (9), 095009.
- Baicker, Katherine, Sarah L. Taubman, Heidi L. Allen, Mira Bernstein, Jonathan H. Gruber, Joseph P. Newhouse, Eric C. Schneider, Bill J. Wright, Alan M. Zaslavsky, Amy N. Finkelstein, and Oregon Health Study Group**, “The Oregon Experiment—Effects of Medicaid on Clinical Outcomes,” *New England Journal of Medicine*, 2013, 368 (18), 1713–1722.
- Bailey, Martha J., Connor Cole, Morgan Henderson, and Catherine Massey**, “How Well Do Automated Linking Methods Perform? Lessons from U.S. Historical Data,” *Journal of Economic Literature*, 2020, 58 (4), 997–1044.
- Barnes, Leighton Pate, Yanjun Han, and Ayfer Özgür**, “Lower Bounds for Learning Distributions under Communication Constraints via Fisher Information,” *Journal of Machine Learning Research*, 2020, 21 (236), 1–30.
- Basu, D.**, “On Partial Sufficiency: A Review,” *Journal of Statistical Planning and Inference*, 1978, 2 (1), 1–13.
- Blackwell, David**, “Equivalent Comparisons of Experiments,” *The Annals of Mathematical Statistics*, 1953, 24 (2), 265–272.
- Chen, Qihui and Zheng Fang**, “Improved Inference on the Rank of a Matrix,” *Quantitative Economics*, 2019, 10 (4), 1787–1824.
- Deshpande, Manasi and Yue Li**, “Who Is Screened Out? Application Costs and the Targeting of Disability Programs,” *American Economic Journal: Economic Policy*, 2019, 11 (4), 213–248.
- Elfving, Gustav**, “Optimum Allocation in Linear Regression Theory,” *The Annals of Mathematical Statistics*, 1952, 23 (2), 255–262.
- Fang, Zheng and Andres Santos**, “Inference on Directionally Differentiable Functions,” *The Review of Economic Studies*, 2019, 86 (1), 377–412.

- Feller, William**, *An Introduction to Probability Theory and Its Applications*, 3 ed., Vol. 1, New York: Wiley, 1968.
- Finkelstein, Amy, Sarah Taubman, Bill Wright, Mira Bernstein, Jonathan Gruber, Joseph P. Newhouse, Heidi Allen, Katherine Baicker, and Oregon Health Study Group**, “The Oregon Health Insurance Experiment: Evidence from the First Year,” *The Quarterly Journal of Economics*, 2012, 127 (3), 1057–1106.
- Firpo, Sergio, Antonio F. Galvao, and Thomas Parker**, “Uniform Inference for Value Functions,” *Journal of Econometrics*, 2023, 235 (2), 1680–1699.
- Giannella, Eric, Tatiana Homonoff, Gwen Rino, and Jason Somerville**, “Administrative Burden and Procedural Denials: Experimental Evidence from SNAP,” *American Economic Journal: Economic Policy*, 2024, 16 (4), 316–340.
- Goel, Prem K. and Morris H. DeGroot**, “Comparison of Experiments and Information Measures,” *The Annals of Statistics*, 1979, 7 (5), 1066–1077.
- Hébert, Benjamin and Michael Woodford**, “Rational Inattention When Decisions Take Time,” *Journal of Economic Theory*, 2023, 208, 105612.
- Hong, Han and Jessie Li**, “The Numerical Delta Method,” *Journal of Econometrics*, 2018, 206 (2), 379–394.
- Lee, David S.**, “Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects,” *The Review of Economic Studies*, 2009, 76 (3), 1071–1102.
- Lin, Shurong and Eric D. Kolaczyk**, “Data Privacy for Record Linkage and Beyond,” Working Paper 32940, National Bureau of Economic Research 2024.
- Louis, Thomas A.**, “Finding the Observed Information Matrix When Using the EM Algorithm,” *Journal of the Royal Statistical Society: Series B*, 1982, 44 (2), 226–233.
- Miao, Jianjun, Jieran Wu, and Eric R. Young**, “Multivariate Rational Inattention,” *Econometrica*, 2022, 90 (2), 907–945.
- National Bureau of Economic Research**, “Oregon Health Insurance Experiment Public Use Data,” NBER Research Data 2026. Accessed August 29, 2026.
- Pomatto, Luciano, Philipp Strack, and Omer Tamuz**, “The Cost of Information: The Case of Constant Marginal Costs,” *American Economic Review*, 2023, 113 (5), 1360–1393.
- Sagnol, Guillaume**, “Computing Optimal Designs of Multiresponse Experiments Reduces to Second-Order Cone Programming,” *Journal of Statistical Planning and Inference*, 2011, 141 (5), 1684–1708.
- Tahamont, Sarah, Zubin Jelveh, Aaron Chalfin, Shi Yan, and Benjamin Hansen**, “Dude, Where’s My Treatment Effect? Errors in Administrative Data Linking and the Destruction of Statistical Power in Randomized Experiments,” *Journal of Quantitative Criminology*, 2021, 37 (3), 715–749.